

MATRICES

1, Let $A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$ then $A^T = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}$

$$\text{Let } P = \frac{1}{2}(A + A^T) = \frac{1}{2} \begin{bmatrix} 6 & 6 \\ 6 & -2 \end{bmatrix}$$

→ clearly $P^T = P$

∴ P is symmetric

$$\text{Let } Q = \frac{1}{2}(A - A^T) = \frac{1}{2} \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$$

clearly $Q^T = -Q$

∴ Q is skew-symmetric

$$\text{Now } P + Q = \frac{1}{2} \begin{bmatrix} 6 & 10 \\ 2 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} = \underline{\underline{A}}$$

2, $A^2 = A \cdot A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$

$$= \begin{bmatrix} 9-8 & -6+4 \\ 12-8 & -8+4 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}$$

Given $A^2 = KA - 2I$

$$A^2 + 2I = KA$$

$$\begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3K & -2K \\ 4K & -2K \end{bmatrix}$$

$$\begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 3K & -2K \\ 4K & -2K \end{bmatrix}$$

$$\therefore 3K = 3$$

$$K = \underline{\underline{1}}$$

$$3, \begin{bmatrix} 3x & 3y \\ 3z & 3w \end{bmatrix} = \begin{bmatrix} x+4 & x+y+6 \\ -1+2+w & 2w+3 \end{bmatrix}$$

$$\begin{array}{l|l|l} 3x = x+4 & 3y = x+y+6 & 3w = 2w+3 \\ 2x = 4 & 2y = 8 & w = 3 \\ x = 2 & y = 4 & \end{array}$$

$$3z = -1+2+w$$

$$2z = 2$$

$$z = \underline{\underline{1}}$$

$$4, \text{ Let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

$$a_{11} = 2-1=1 \quad a_{12} = 2-2=0 \quad a_{13} = 2-3=-1$$

$$a_{21} = 4-1=3 \quad a_{22} = 4-2=2 \quad a_{23} = 4-3=1$$

$$\therefore A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 1 \end{bmatrix}$$

$$5, x+y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \text{ --- (1) } \quad x-y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \text{ --- (2)}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow 2x = \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix}$$

$$\therefore x = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$$

$$\textcircled{1} - \textcircled{2} \Rightarrow 2Y = \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix}$$

$$Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

$$6, AB = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -6 & 12 \\ 4 & 12 & -24 \\ 5 & 15 & -30 \end{bmatrix}$$

$$(AB)^T = \begin{bmatrix} -2 & 4 & 5 \\ -6 & 12 & 15 \\ 12 & -24 & -30 \end{bmatrix} \quad \text{--- (1)}$$

$$B^T A^T = \begin{bmatrix} 1 \\ 3 \\ -6 \end{bmatrix} \begin{bmatrix} -2 & 4 & 5 \end{bmatrix} = \begin{bmatrix} -2 & 4 & 5 \\ -6 & 12 & 15 \\ 12 & -24 & -30 \end{bmatrix} \quad \text{--- (2)}$$

$$\therefore (AB)^T = B^T A^T$$

$$7, O(AB) = 2 \times 2$$

8, Skew-symmetric

$$9, A^2 = A \cdot A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

$$10, \text{ Let } A = IA$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A \quad R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} A \quad R_1 \rightarrow R_1 - 3R_2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix} A$$

$$\therefore \underline{\underline{\tilde{A}^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}}}$$

Answer